नमामि गंगे

AMITY UNIVERSITY

RIVERATHON

INNOVATE TO REVIVE, RESTORE, REJUVENATE RIVER

A National Level River Hackathon !

RIVERATHON 1.0

(Innovate to Revive, Restore, Rejuvenate River)

IDEA SUBMISSION

INTRODUCTION
(Team & Idea Overview)

- **Team Name: RIVERDOT**
- **Team Details:**

Name	Email ID	Contact Number	Institution	Team Leader (Yes/No)
REEDDHIJIT DEB	reeddhijitdeb@gmail.com	8777547075	DR DY PATIL INST. OF TECH, PIMPRI	YES
TANISHA PHUKAN	tanishaphukan.1705@gmail.com	8638906691	DR DY PATIL INST. OF TECH, PIMPRI	NO

- **Idea Title: Sedimentation & Erosion Hotspot Predictor for River Systems - (Jal kavaach)**
- **Theme: Sediment Management**

PROBLEM STATEMENT

TO DEVELOP A PREDICTIVE TOOL FOR IDENTIFYING SEDIMENTATION AND EROSION HOTSPOTS IN RIVERS USING LIDAR DATA.

- **Real-World Problem**
  - Human activities & climate change disrupt natural sediment flow.
  - Causes flooding, land loss, infrastructure & habitat damage.

- **Why It Matters**
  - Affects water capacity, ecosystems & public safety.
  - Early detection = smarter, cost-effective interventions.

- **Relevant Data**
  - 20% reservoir capacity lost due to sedimentation (CWC)
  - 60% river infra failures linked to erosion (NIH)

- **How LiDAR Helps**
  - High-res terrain & river profile mapping.
  - Tracks changes in elevation & slope.
  - Enables precise, predictive erosion/sedimentation analysis.

PROPOSED SOLUTION

- **Explain the proposed idea/solution.**
  - To build a prediction tool that uses LiDAR data, river flow information, and terrain features to identify areas in rivers that are prone to sedimentation and erosion.
- **How your solution addresses the problems and challenges described in the previous slide?**
  - Analyzes river slope, flow speed, and channel shape.
  - Detects where erosion (bank wearing) or sediment build-up is likely.
  - Helps decision-makers before take action serious damage happens.
- **What makes it unique or innovative?**
  - Uses high-resolution LiDAR for accurate riverbed and bank mapping.
  - Can simulate changes during floods, droughts, etc.
  - Generates easy-to-read hotspot maps to guide river management.

TARGET USERS & IMPACT

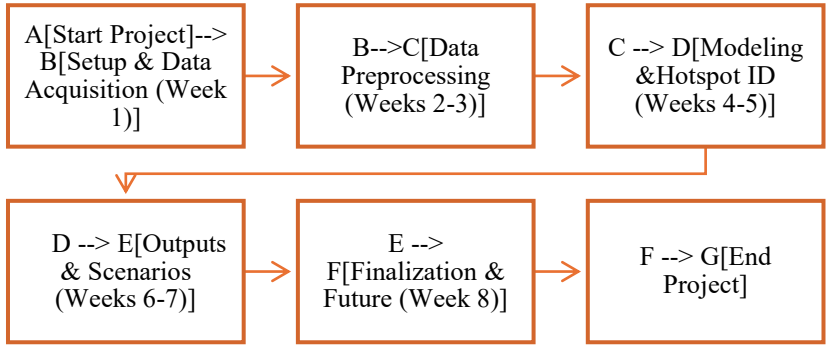
- **Primary users/beneficiaries**
  - WRD, CWC, river authorities.
  - Environmental agencies and conservation groups.
  - Urban planners and civil engineers.
  - Disaster response teams.
  - Local riverine communities.
- **How the solution helps them**
  - Predicts erosion/sedimentation zones early.
  - Minimizes infrastructure damage.
  - Enables focused, cost-effective action.
  - Enhances flood preparedness.
- **Potential social, economic, or environmental impact**
  - Social: Safer, more resilient communities.
  - Economic: Reduced maintenance & recovery costs.
  - Environmental: Healthier rivers and ecosystems.

TOOLS & TECHNOLOGY STACK

- **Tools, platforms, or programming languages to be used**
  - GIS Software, Python
  - Python Libraries (NumPy, SciPy, Pandas, GeoPandas, PyGEOS, Matplotlib, Seaborn)
- **Technology Stack**
  - Frontend- Folium, Plotly Express, Leaflet.js (with Python backend)
  - Backend- Python, QGIS Software
  - Python libraries-
    - Geospatial data analysis- GDAL/OGR, Rasterio, Fiona
    - Numerical computation- NumPy, SciPy
    - Time series data analysis- Pandas
    - Vector spatial analysis- GeoPandas, PyGEOS
- **LiDAR Use**
  - It will provide accurate representation of river morphology, which is crucial for detailed analysis of river systems by studying river features such as slope, curvature, roughness, and precise channel cross-sections. This direct input into models allows for more accurate calculations of flow dynamics and shear stress, thereby improving the prediction of erosion and deposition hotspots.

IMPLEMENTATION PLAN

- **Methodology**
  - **Data Preparation:** Load and clean all spatial and temporal data and convert it to a usable format.
  - **Flow & Sediment Modeling:** Use simplified equations to estimate river flow velocity, depth, and force of water on riverbed.
  - **Risk Assessment:** To develop specific indices that combine shear stress with river characteristics to identify areas prone to erosion.
  - **Hotspot Identification:** Identify thresholds for these indices and use spatial analysis to group high-risk areas into "hotspots".
  - **Scenario Prediction:** Allow users to view how these hotspots change under different hypothetical river flow scenarios.
- **Plan for developing the solution with timeline**



- **Team's preparedness**
  - Reeddhijit Deb- Frontend, Python ,GIS Software.
  - Tanisha Phukan- Python Libraries, Data Acquisitions.

FEASIBILITY & CHALLENGES

- **Feasibility**
  - A small team of 2 people with a mix of skills in Python programming, basic data analysis, and web development (even just front-end) is enough to tackle the MVP.
- **Potential challenges and risks**
  - Tech setup- environment setup, trying to make the hydrological model too accurate.
  - Data hunting and CRS(Coordinate reference system mismatch) common in GIS software.
- **Strategies for overcoming these challenges**
  - Pre-built environment setup.
  - Work with data subset.
  - Define a micro MVP first.
- **What makes it innovative**
  - It's an interactive tool that instantly shows how floods change erosion risks.
  - It makes complex analysis easy for non-experts to use it on a simple website.
  - It helps predict and prevent damage instead of just reacting on it after it happens.

EXPECTED OUTCOME

- **Demonstration**
  - The problem statement.
  - The solution- Shows a live map.
  - WoW moment- After simulation it will show the risk zone areas.
  - Impact- It can be used to prevent floods during disaster season.
- **Potential for scalability of data**
  - For larger data- Use a spatial database like PostGIS.
  - For high performance- Move all heavy processing to the cloud (AWS, Google Cloud).
  - For better tech. - Divide the application into a separate backend API and a good frontend.
- **Future scope**
  - Add Machine Learning to make it more accurate predictions.
  - Connect to live IoT sensors to show the real-time river conditions.
  - Suggest solutions and estimate costs for fixing high-risk areas.

ADDITIONAL INFO & DECLARATION

- **Prototype website-**

  https://reedddeb-12.github.io/jalkavaach/

- **Github repository-**

  https://github.com/Reedddeb-12/jalkavaach.git

"We declare this idea is original and created by our team."